CM617 | COURSE PROJECT

Lifecycle Carbon Emissions: Electric vs. Fuel Vehicles

Carbon payback, electricity grids, vehicle type and adoption

Team: Manasvi Kadam (23B1300), Riddhi Gupta (23B3989), Anjali Joshi (23B3949), Shreya Nigam (23B1258)

Overview

Four complementary analyses compare electric vehicle (EV) and internal-combustion-engine (ICE) lifecycle emissions: global lifecycle and electricity-grid trends; Indian scooter, car and bus comparisons; three vehicle types across nine electricity grids; and break-even results for two-wheelers, sedans and SUVs in Norway, France, the United States and India. Data cover EV sales and stock, grid carbon intensity, vehicle efficiency, carbon dioxide (CO_2), gross domestic product (GDP), energy mix and urbanisation.

Headline figures: Over 150,000 kilometres, a petrol car produces 45.94 tonnes of CO_2; an EV produces 12.98 tonnes (Norway), 25.42 tonnes (United States) and 35.39 tonnes (India). Typical reported carbon payback is under 3 years.

Lifecycle, Grid and Fleet Findings

- **Manufacturing/use:** EV manufacture emits about 12 tonnes of CO_2 versus 7 tonnes for a petrol car in the baseline, but lower emissions during driving repay the difference.
- **Grid intensity since 2000 (grams of CO_2 per kilowatt-hour):** United States: 607.5→384.4; China: 783.5→525.3; India: 740.0→670.1; Norway: 25.8→28.1.
- **Renewable share, 2013→2025:** Germany: 24.1→59.1%; China: 20.1→37.0%; United States: 12.8→25.6%; India: 16.0→24.1%; Norway: 98.0→99.0%.
- **Vehicle size/materials (India model):** Petrol SUV/sedan: 0.259/0.224 kilograms of CO_2 per kilometre; EV SUV/sedan: 0.163/0.150 kilograms of CO_2 per kilometre. Per 1 megawatt-hour of battery capacity, small sedans are estimated to save 282,955 kilograms of CO_2 versus 213,040 kilograms for large SUVs.
- **Grid crossover:** One model estimates the electricity-grid crossover at 1,115 grams of CO_2 per kilowatt-hour for petrol and 1,130 grams for diesel; another gives 1,286–1,631 grams per kilowatt-hour depending on the vehicle. These estimates use different assumptions.
- **Adoption/urbanisation:** Reported EV counts: China: 360 million; United States: 36 million; South Korea: 15.8 million; India: 14.5 million; Japan: 5.4 million. India: 35.7% urbanisation and 10.2 EVs per 1,000 people; China: 66.3% and 252.6 EVs per 1,000 people. Association with wealth or urbanisation does not prove causation.
- **Carbon debt:** Replacing 1 million petrol cars adds 5.00 million tonnes of CO_2 during manufacturing; the net difference is an increase of 2.68 million tonnes in year 1 and 0.35 million tonnes in year 2, followed by savings of 1.97, 4.29 and 6.61 million tonnes in years 3–5.

India: Scooter, Car and Bus

Pairs: Ola S1 Air/Honda Activa 6G; Tata Nexon EV Max/petrol Nexon; 12-metre Switch EiV 12/diesel city bus. Assumptions: battery production: 68 kilograms of CO_2-equivalent emissions per kilowatt-hour; electricity grid: 670 grams of CO_2 per kilowatt-hour; charging efficiency: 90%; transmission and distribution loss: 17%. EV driving emissions include electricity use and losses; ICE emissions include additional emissions from fuel production.

	Scooter	Car	Bus
EV use (grams of CO_2 per kilometre)	28.3	127.5	944
ICE use (grams of CO_2 per kilometre)	58.0	209.4	1,054
Extra EV manufacturing (kilograms of CO_2)	204	2,754	19,244
Break-even distance (kilometres)	∼6,900	∼33,600	∼175,000

Scooters repay manufacturing emissions fastest because their batteries are small; buses have a larger manufacturing penalty but high annual use. These vehicle pairs and assumptions differ from other analyses and should not be combined directly.

Nine-Grid Comparison and Sensitivity

A separate model compares TVS iQube, Tata Tiago and Tata Harrier EV with petrol counterparts in India, Poland, China, Japan, the United States, Germany, the United Kingdom, France and Norway. Battery footprint: 70 kilograms of CO_2-equivalent emissions per kilowatt-hour; charging loss: 10%; grid losses: 17.7% in India and 6% elsewhere. In 2025, grid intensity ranges from about 670 grams of CO_2 per kilowatt-hour in India to 28 grams in Norway; India's effective charging electricity intensity is about 814 grams per kilowatt-hour.

- **Break-even:** Approximately 4,200 kilometres (Norway scooter) to 36,100 kilometres (India mid-size SUV). Time: scooters 0.5–0.9 years, hatchbacks 0.7–1.8 years, SUVs 1.4–3.0 years.
- **Mileage sensitivity:** At 60%, 100% and 150% of standard annual distance driven, the break-even distance is unchanged but the time changes. India baseline times: scooter 0.9 years, hatchback 1.8 years, SUV 3.0 years.
- **Battery sensitivity in India (50–100 kilograms of CO_2-equivalent emissions per kilowatt-hour):** break-even distance: 5,600–11,000 kilometres (scooters), 15,000–30,000 kilometres (hatchbacks), 26,000–52,000 kilometres (SUVs). Highway-efficient petrol SUVs: 80,000–114,000 kilometres; extreme estimate using ARAI efficiency: about 950,000 kilometres.
- **Reduction in driving emissions:** France/Norway: 97–98%; United Kingdom/Germany: 74–87%; United States/Japan: 63–76%; China/Poland: 54–68%; India: 41–53%.

Four-Country Break-Even Model

Assumed extra manufacturing emissions: two-wheeler: 312.5 kilograms of CO_2; sedan: 4,250 kilograms; SUV: 6,500 kilograms. Formula:

$$D_{\mathrm{BE}} = \frac{B \times f_{\mathrm{batt}}}{e_{\mathrm{petrol}} - e_{\mathrm{EV}}}, \qquad t_{\mathrm{BE}} = \frac{D_{\mathrm{BE}}}{\text{annual distance driven}},$$

$$e_{\mathrm{EV}} = \frac{c_{\mathrm{battery}}}{\eta_{\mathrm{charge}}} \times \frac{I_{\mathrm{grid}}}{1 - L_{\mathrm{grid}}}, \quad e_{\mathrm{petrol}} = \frac{2{,}935\ \mathrm{g/L}}{\mathrm{km/L}}.$$

Here, D_{BE} is in km and t_{BE} in years; B is battery capacity (kWh), f_{batt} the battery footprint (g CO_2e/kWh), c_{battery} the battery-to-wheel consumption (kWh/km), η_{charge} the charging efficiency, I_{grid} the grid carbon intensity (g/kWh generated) and L_{grid} the grid loss share. If $e_{\mathrm{EV}} \geq e_{\mathrm{petrol}}$, the EV never breaks even.

Type	Country	Distance (km)	Time (years)	Net savings (%)
2-wheel	Norway	6,273	0.8	88.6%
	France	6,427	0.8	86.3%
	United States	8,676	1.1	62.7%
	India	13,938	1.7	37.2%
Sedan	Norway	17,608	1.2	82.1%
	France	18,067	1.2	79.9%
	United States	24,880	1.7	56.1%
	India	41,954	2.8	30.4%
SUV	Norway	20,425	1.1	83.0%
	France	20,930	1.2	80.9%
	United States	28,271	1.6	58.1%
	India	45,492	2.5	33.5%

Norway (hydroelectric power) and France (nuclear power) reach payback fastest; the United States is intermediate. India takes roughly 2.2–2.4 times Norway's payback time, but reports 30–37% lifetime savings. SUVs have larger manufacturing penalties but can save more fuel; two-wheelers pay back fastest in absolute terms.

Adoption and Costs

Global 2025 EV sales: cars: 21 million (25% share); two- and three-wheelers: 11 million (15%); trucks: 440,000 (9%); vans: 430,000 (4%); buses: 70,000 (5%). China leads by sales share. India: two- and three-wheelers: 2.1 million (10%); cars: 170,000 (4%). IEA projections for India, 2025→2035: annual car sales: 0.17→0.43 million; car stock: 0.41→3.5 million; annual two- and three-wheeler sales: 2.1→4.9 million; stock: 8.7→44 million.

Average battery-electric vehicle (BEV) price is about $23,000 versus $15,000 for an ICE vehicle (+53%). City cars: $10,000 versus $9,000 (+12%); small SUVs/pick-ups: $20,000 versus $16,000 (+25%); medium cars: +189%; large cars: +81%; large SUVs: +154%. India public chargers (2025): 55,000 slow (63%), 31,000 fast (35%), 2,000 ultra-fast (2%); ultra-fast charging capacity remains limited.

Conclusion

Conclusion: EVs generally repay extra manufacturing emissions within a few years and save carbon thereafter. For India, prioritise two- and three-wheelers and affordable small cars alongside grid decarbonisation; large EVs on coal-heavy grids need careful sensitivity analysis.

Graphs and Visual Analysis

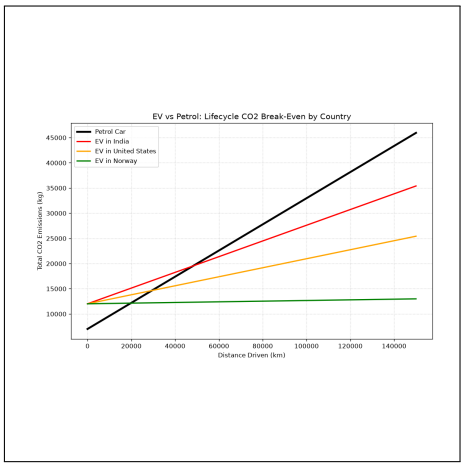


Figure 1: Breakeven Point

EV vs ICE Vehicle: Lifecycle CO2 Break-Even by Country

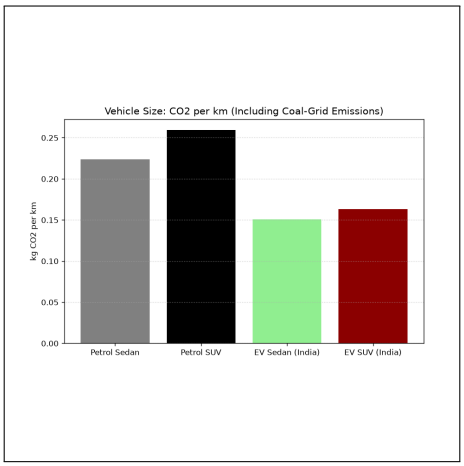


Figure 2: Vehicle Size

Vehicle Size: CO2 per km (Including Coal-Grid Emissions)

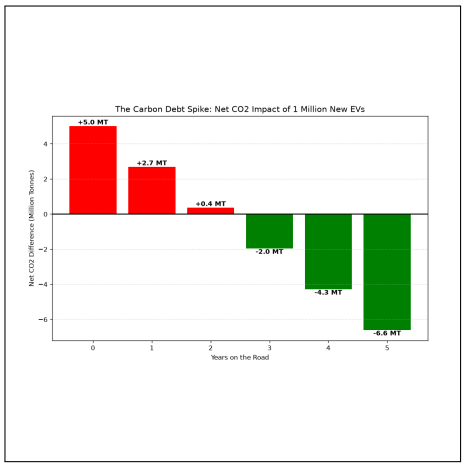


Figure 3: 1 Million ICE Replacement Effect

The Carbon Debt Spike: Net CO2 Impact of 1 Million New EVs

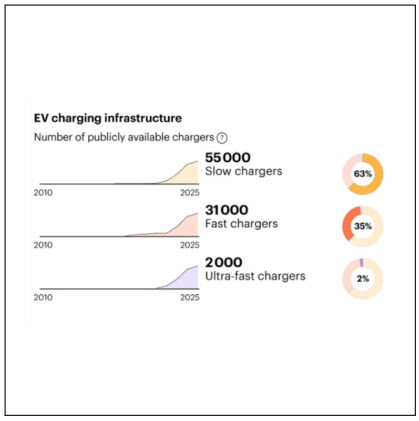


Figure 4: EV Charging Infrastructure

Number of publicly available chargers.

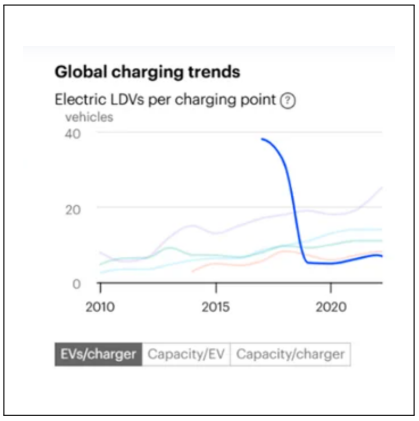


Figure 5: Global Charging Trends

Electric light-duty vehicles per public charging point.

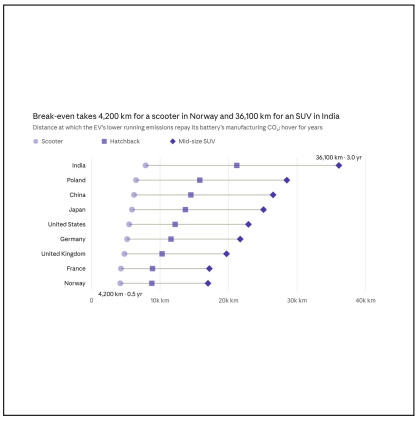


Figure 6: Break-Even point

Distance at which the EV's lower running emissions repay its battery's manufacturing CO2

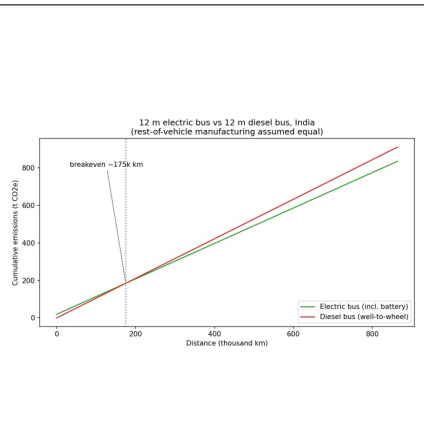


Figure 7: Comparison

Electric Bus(Switch EiV 12) vs Diesel Bus

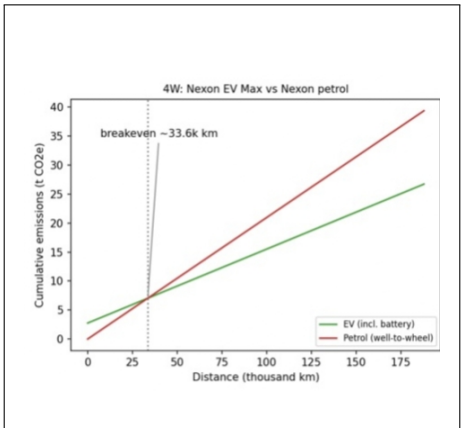


Figure 8: Comparison

4-wheeler: Tata Nexon EV vs Petrol

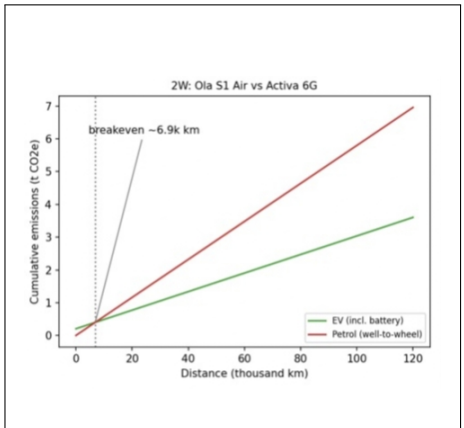


Figure 9: Comparison

2 wheeler: Ola electric vs Activa 6G